

# pyDMS

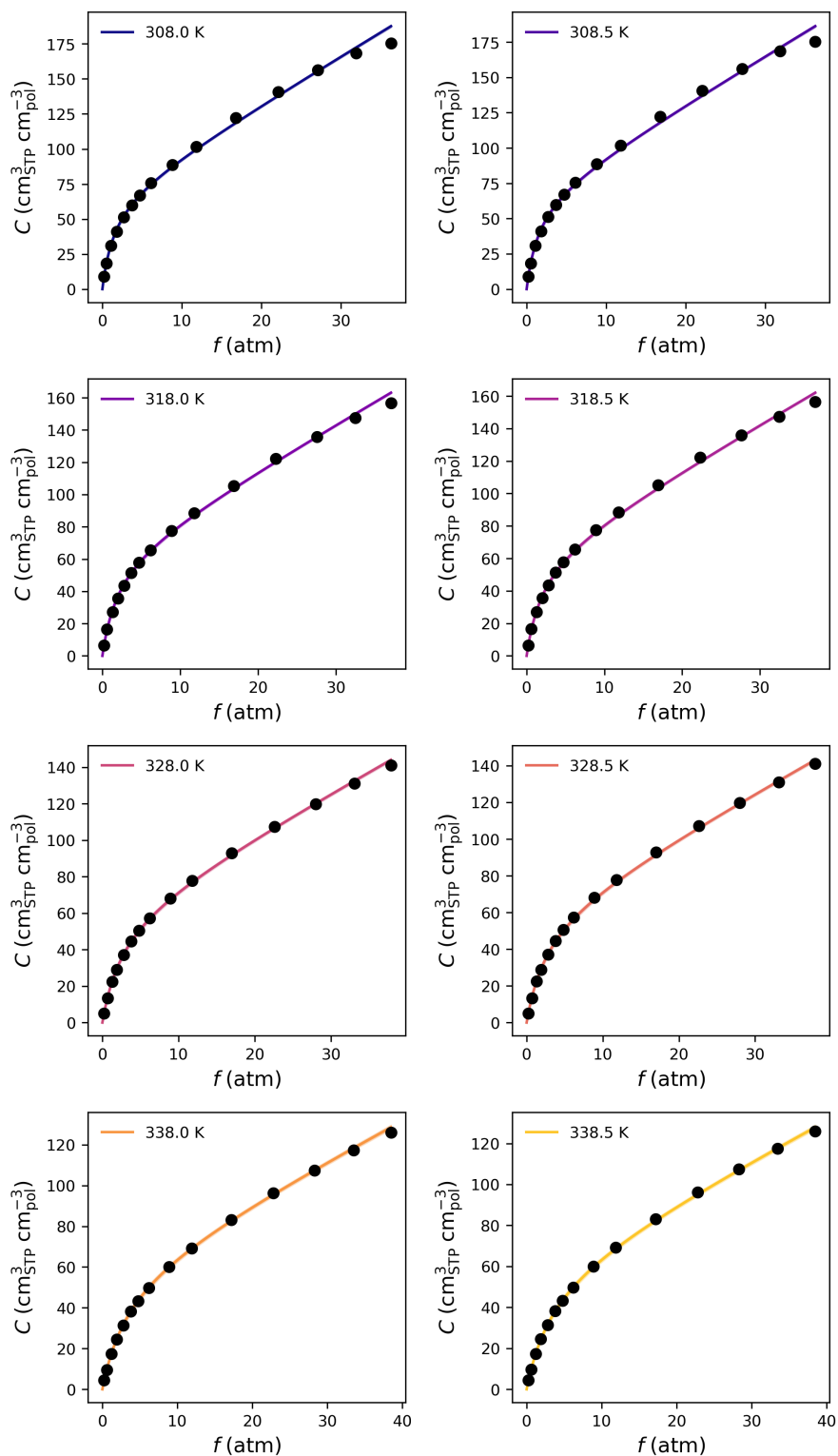
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2025-06-13 10:35:47 Eastern Daylight Time

## DMS Parameters

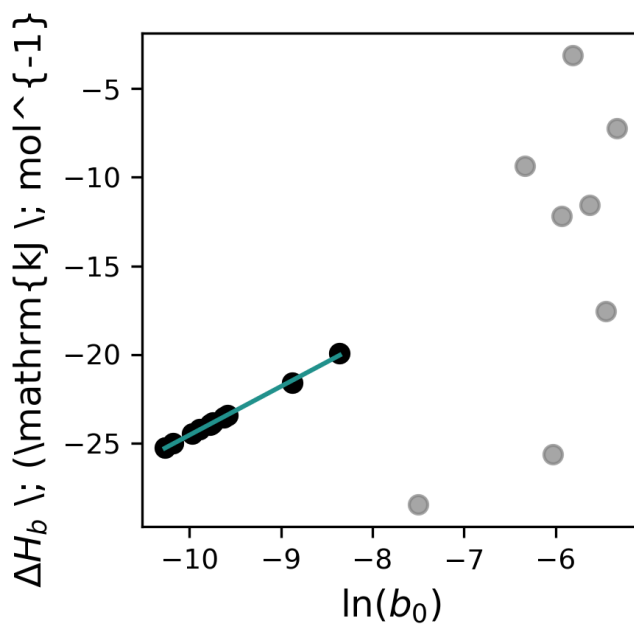
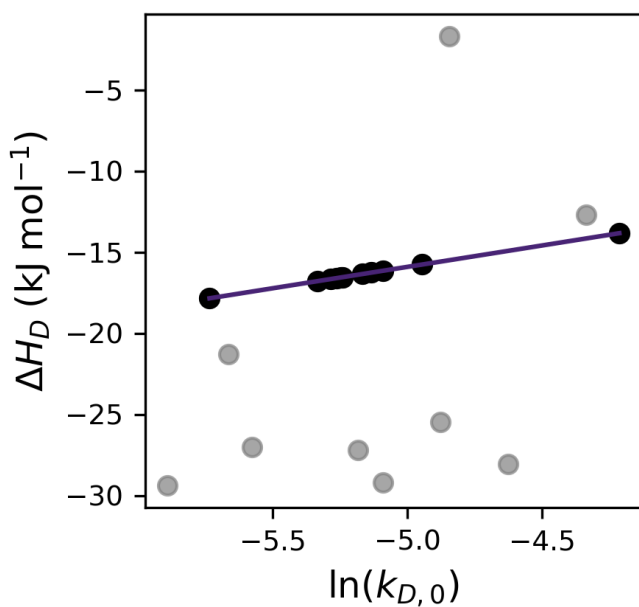
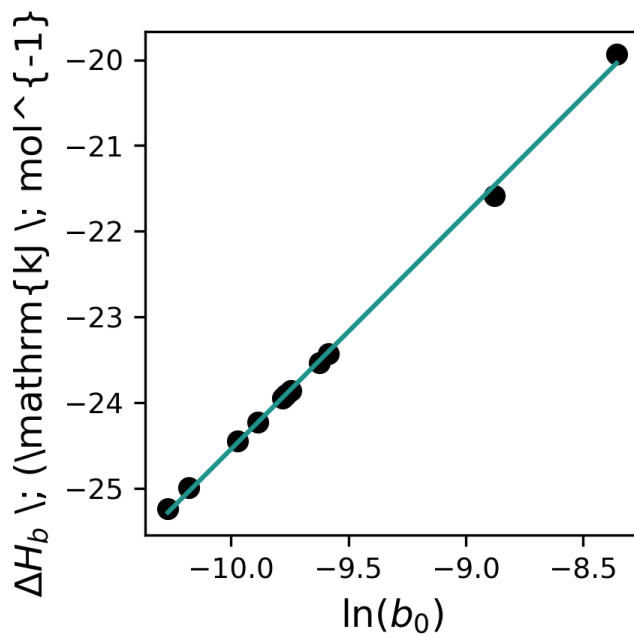
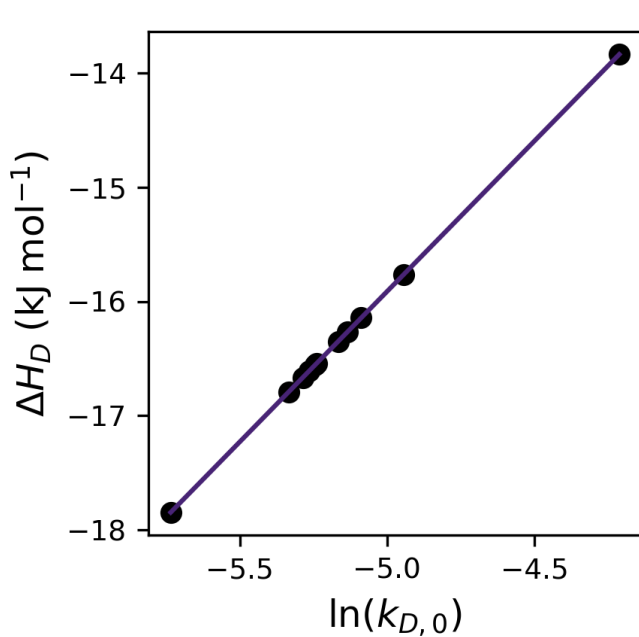
| Temperature | Parameter | Result $\pm$ Error                             |
|-------------|-----------|------------------------------------------------|
| 308.0 K     | $k_D$     | $3.3863029898295274 \pm 0.03575736274332727$   |
|             | $C'_H$    | $67.42639509874338 \pm 0.687694173188234$      |
|             | $b$       | $0.6526145073061795 \pm 0.09780411741985409$   |
| 308.5 K     | $k_D$     | $3.351228873561245 \pm 0.03386081722797417$    |
|             | $C'_H$    | $67.4263950987432 \pm 0.6921034085096665$      |
|             | $b$       | $0.6427300713642302 \pm 0.0963202447197424$    |
| 318.0 K     | $k_D$     | $2.766896893865147 \pm 0.014242145010207596$   |
|             | $C'_H$    | $63.83797770307673 \pm 0.7477418795396744$     |
|             | $b$       | $0.4853497733145708 \pm 0.07270816066789526$   |
| 318.5 K     | $k_D$     | $2.740002433113007 \pm 0.01439475027241707$    |
|             | $C'_H$    | $63.837977703076746 \pm 0.7525763515111947$    |
|             | $b$       | $0.4784501748509722 \pm 0.07167363148524088$   |
| 328.0 K     | $k_D$     | $2.2888097818792246 \pm 0.026494478936010247$  |
|             | $C'_H$    | $61.34795506469597 \pm 0.8382277343097684$     |
|             | $b$       | $0.3675314579191949 \pm 0.05505005500727475$   |
| 328.5 K     | $k_D$     | $2.2678911167732663 \pm 0.0271960200068049$    |
|             | $C'_H$    | $61.34795506469606 \pm 0.8437325346859197$     |
|             | $b$       | $0.36261811634585667 \pm 0.054314021619732845$ |
| 338.0 K     | $k_D$     | $1.9147019339399491 \pm 0.038331484227305895$  |
|             | $C'_H$    | $59.9037595494776 \pm 0.9750915877526438$      |
|             | $b$       | $0.28293051120018253 \pm 0.04238061043341015$  |
| 338.5 K     | $k_D$     | $1.898217453721392 \pm 0.03879926227750595$    |
|             | $C'_H$    | $59.90375954947754 \pm 0.9813179823640752$     |
|             | $b$       | $0.279367099963692 \pm 0.04184715697002851$    |

## Sorption Isotherms

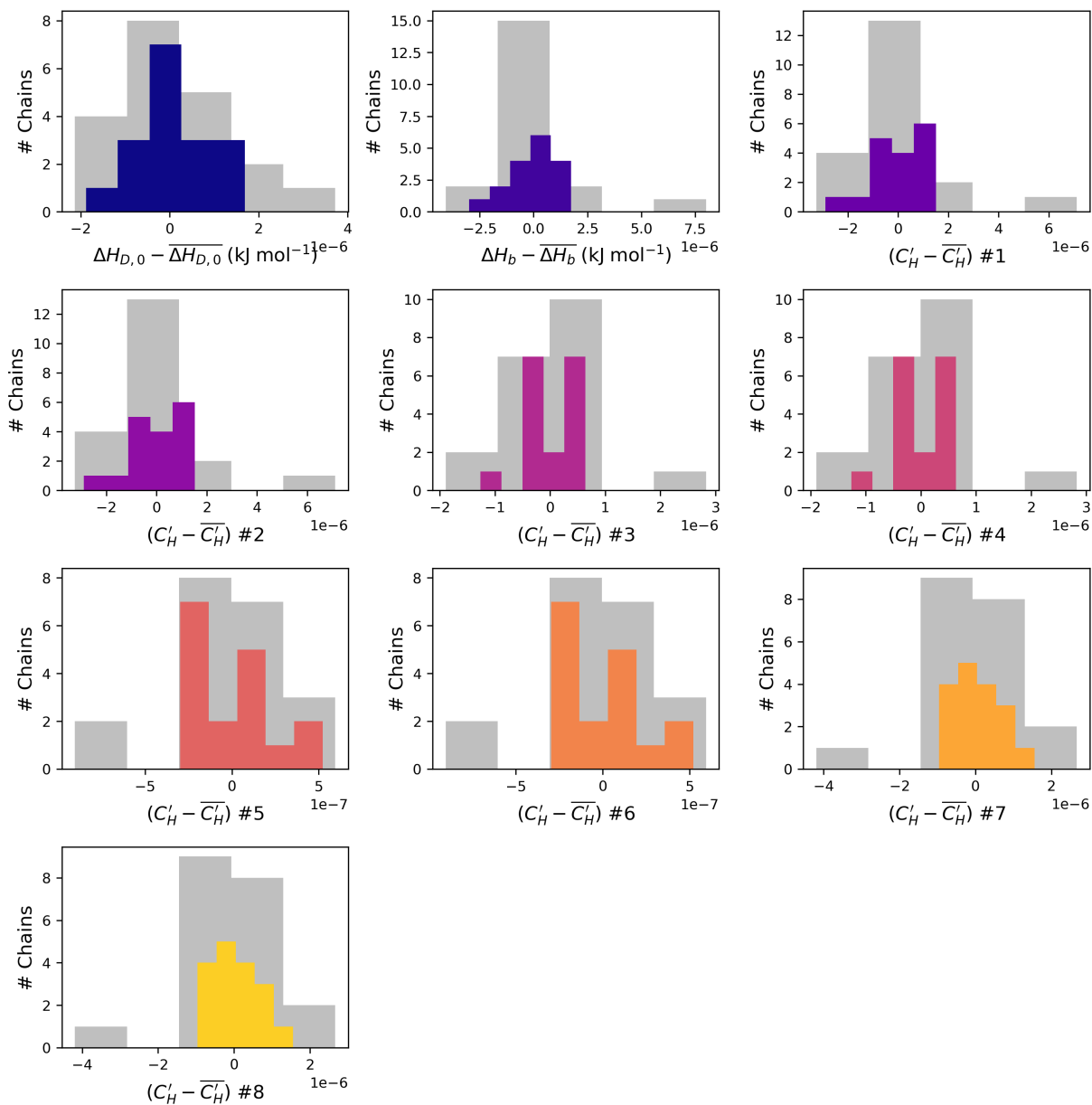


## Linear Free Energy Relationships

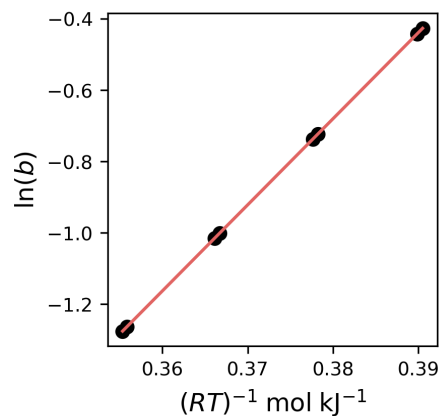
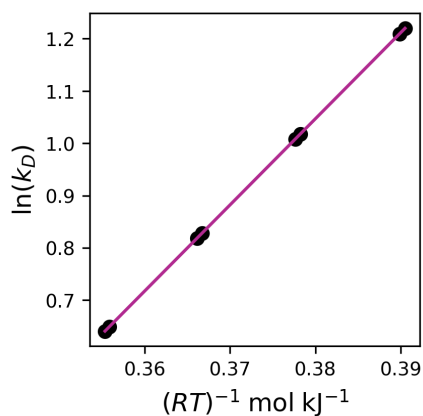
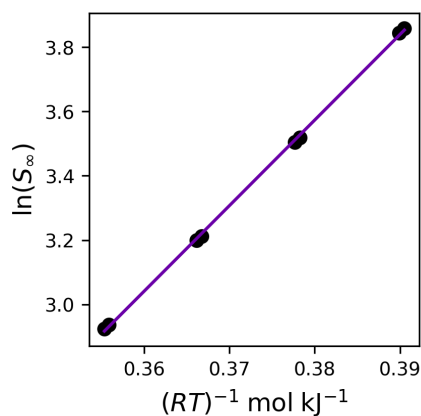
|                           |                                              |
|---------------------------|----------------------------------------------|
| Heat of Henry Sorption    | $\Delta H_D = -2.7294 + 2.6364 \ln(k_{D,0})$ |
| Heat of Langmuir Sorption | $\Delta H_b = 2.895 + 2.7438 \ln(b_0)$       |



## van't Hoff Relationships



## Energetics



|                       | Entropic Prefactor ( $S_{i,0}$ )        | $\Delta H_i$ (kJ/mol)     |
|-----------------------|-----------------------------------------|---------------------------|
| i = Infinite Dilution | $1.455\text{e-}03 \pm 1.138\text{e-}04$ | $-26.594293 \pm 0.209829$ |
| i = Langmuir          | $5.493\text{e-}03 \pm 1.666\text{e-}16$ | $-16.449993 \pm 0.825950$ |
| i = Henry             | $5.310\text{e-}05 \pm 3.771\text{e-}18$ | $-24.112981 \pm 0.190649$ |

## Analysis Settings

| Parameter                        | Value           |
|----------------------------------|-----------------|
| $\Delta H_D$ Initial Guess Range | [-1, -30]       |
| $\Delta H_b$ Initial Guess Range | [-1, -30]       |
| $k_{D,0}$ Initial Guess Range    | [0.001, 0.01]   |
| $b_0$ Initial Guess Range        | [0.0001, 0.005] |
| $\Delta H_D$ Solver Bounds       | [-50, 0]        |
| $\Delta H_b$ Solver Bounds       | [-50, 0]        |
| $k_{D,0}$ Solver Bounds          | [0, None]       |
| $b_0$ Solver Bounds              | [0, None]       |
| $C'_H$ #1                        | [ 0 150]        |
| $C'_H$ #2                        | [ 0 150]        |
| $C'_H$ #3                        | [ 0 150]        |
| $C'_H$ #4                        | [ 0 150]        |
| $C'_H$ #5                        | [ 0 150]        |
| $C'_H$ #6                        | [ 0 150]        |
| $C'_H$ #7                        | [ 0 150]        |
| $C'_H$ #8                        | [ 0 150]        |
| Trials                           | 20              |
| LFER Solver                      | SLSQP           |
| maxiter (LFER solver)            | 1000            |
| van't Hoff Solver                | SLSQP           |
| maxiter (van't Hoff solver)      | 1000            |
| ftol (SLSQP)                     | 1e-12           |
| xtol (trust-constr)              | 1e-12           |
| gtol (trust-constr)              | 1e-12           |

## Questions/Comments/Concerns?

Documentation available through ReadTheDocs (\*)

Please reach out to Brandon C. Tapia via email ([bctapia@mit.edu](mailto:bctapia@mit.edu)) or through the pyDMS GitHub page

## Code Availability

pyDMS can be downloaded via:

- GitHub (source code): `git clone \*\*insert link`
- pip: `pip install \*\*insert name`
- Anaconda: `conda install \*\*insert name`

## Citation

If you used pyDMS in any presented or published work please cite:

## License

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